

AstraPay — Payment Contribution Profitability

Task 1: Executive Problem Framing

Executive Core Question

"Why is payment contribution profit per successful transaction falling even though payment volume is increasing, and what should AstraPay change in the next 60 days?"

SMART Objective

Deliver a 5-day diagnostic across AstraPay's India, Singapore, and UAE regional books (Jan–Jun 2025) to:

- Quantify core drivers behind declining unit economics.
 - Isolate true economic shifts from measurement and data-quality artifacts.
 - Deliver a prioritized, executive-approved 60-day action plan featuring named owners and measurable target KPIs.
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Key Performance Indicators (KPIs)

Primary Executive Metric

Contribution Profit per Successful Transaction (USD)

Contribution Profit / Successful Txn = (Merchant Fee Revenue + Realized FX Gain/Loss - Route Processing Cost - Fraud/Chargeback Cost) ÷ Count of Successful Transactions

- **Timing Basis:** `created_at` establishes activity volume; `settlement_date` and settlement FX `rate_type` govern realized revenue, routing costs, and FX conversion.

Supporting Metrics Matrix

Metric	Calculation / Numerator & Denominator	Strategic Focus
Payment Volume	Count of <code>transaction.csv</code> records by <code>created_at</code>	Top-line attempt demand
Success Rate	Successful Attempts ÷ Total Attempts	Operational efficiency & conversion
Gross Payment Value (GPV)	Σ amount (where status = success and amount > 0) in USD	Overall transaction throughput
Processing Cost / Txn	(fixed_fee + variable_fee_pct × amount) ÷ Successful Txns	Network & route cost efficiency
Fraud & Chargeback Cost	Total chargeback amounts attributed to originating month	Risk exposure per successful order
FX Spread Impact	GPV valued at Interbank Rate vs. Realized Settlement Rate	Treasury and conversion drag

Scope & Boundaries

- **In Scope:** All transactions created between 2025-01-01 and 2025-06-30 across India, Singapore, and UAE. Includes all standard channels, routes, merchant tiers, and underlying datasets.
- **Exclusions:**
 - Pending transactions (`status = pending`) due to non-terminal status.
 - Technical zero-amount success records (35 identified capture defects).
 - Unmatched orphan records across settlement, fraud, or chargeback tables.
- **Core Working Assumptions:**
 - AstraPay absorbs 100% of chargeback costs without active merchant liability shift.

- Route processing costs apply according to terms effective at transaction `created_at`.

Key Decisions Enabled

1. **Routing Optimization:** Rebalance traffic routing or renegotiate rates with escalating provider routes.
 2. **Commercial Pricing:** Adjust merchant pricing tiers selectively or hold current contract terms.
 3. **Risk Controls:** Retune or roll back aggressive fraud scoring rules impacting good order flow.
 4. **Data Infrastructure:** Identify critical data pipeline defect fixes required prior to the next reporting cycle.
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MECE Hypothesis Structure

Lens & Stakeholder	Core Hypothesis	Confirmation Criteria	Rejection / Weakening Criteria
1. Commercial / Pricing(<i>CFO</i>)	Merchant fee yield has fallen: Contract concessions or tier shifts reduced overall gross take rate.	Blended fee % of GPV shows continuous decline across overall merchant segments.	Take-rate remains stable once isolated commercial renegotiations are filtered out.
2. Operations & Routing(<i>Head of Payments</i>)	Processing costs spiked: Vendor rate hikes or routing mix shifts drove up unit cost.	Route-level fee increases or volume shifting toward costlier routing options.	Unit route costs and channel volume distribution remain stable over time.
3. Risk & Fraud(<i>Risk Head</i>)	Over-tightened fraud logic: Stricter rules triggered false declines and margin drag.	Decline rates spike significantly alongside lower profitability per attempt.	Higher decline rates are confined to high-risk pools without altering successful unit margins.
4. Data & Measurement(<i>Data Head</i>)	Measurement artifact: Data join errors or FX choices created a false decline.	Margin trend shifts materially depending on FX rate selection or table joining logic.	Unit margin trends remain consistent regardless of methodology or join mechanics.